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# Pluvio Grow — M5 Build

RAIN, WITH A BRAIN.




A 13-plant, weighed-dose plant waterer from M5Stack parts

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# Build a 13-Plant Self-Watering Robot (Open Source) — Pluvio Grow M5

## Intro

I kept killing my plants — so I built a little robot that waters 13 of them, each its own exact amount, measured **by weight**. One arm spins around a carousel, stops at each plant, and doses it from a single reservoir. It runs fully offline on a tiny ESP32, and I made the whole thing source-available so you can build your own.

**What makes it different from a $40 drip timer:** it *weighs* every pour with a load cell, so the cactus gets 15 g and the fern gets 60 g — not "everything for 30 seconds." Rain, with a brain.

- **Difficulty:** intermediate (soldering/crimping, a stepper tweak, 3D printing)
- **Time:** a weekend
- **Cost:** ~**$80–120** in parts for one build (per-unit prices); budget closer to ~**$200–265** at checkout once minimum-order packs and bulk tubing are counted — you'll have spares

## Supplies

### Electronics (DigiKey)

| Part | MPN | Qty | Unit |
|---|---|---|---|
| M5Stack **AtomS3R** dev kit (ESP32-S3 + 0.85″ display) | C126 | 1 | $17.50 |
| M5Stack **Mini Scales Unit** (HX711 load cell) | U177 | 1 | $6.95 |
| M5Stack **Atomic Motion Base** v1.2 (stepper + servo driver) | A090-V12 | 1 | $8.50 |
| **Olimex SM-5VDC** stepper (5-wire; convert to 4-wire **bipolar** — see note) | SM-5VDC | 1 | $1.76 |
| **AS5600** magnetic rotary encoder (arm position) | AliExpress | 1 | — |
| **Diametric magnet, 6 × 2.5 mm** (AS5600 target — presses into the dosing spout on the motor shaft) | generic (AliExpress) | 1 | — |
| 5 V 5 W wall adapter | PSA05A-050QL6-H | 1 | $3.91 |
| USB-A → USB-C cable | DH-20M50053 | 1 | $1.92 |
| Connector housing, 2-pos 2.54 mm | M20-1060200 | 2 | $0.10 |
| Crimp socket, 22–30 AWG gold | M20-1180042 | 2 | $0.09 |

> **Heads-up on DigiKey min-order:** several of these have minimum quantities (e.g. the stepper and adapter ship in packs), so your cart total will exceed the per-unit cost above. Buy one device's worth;

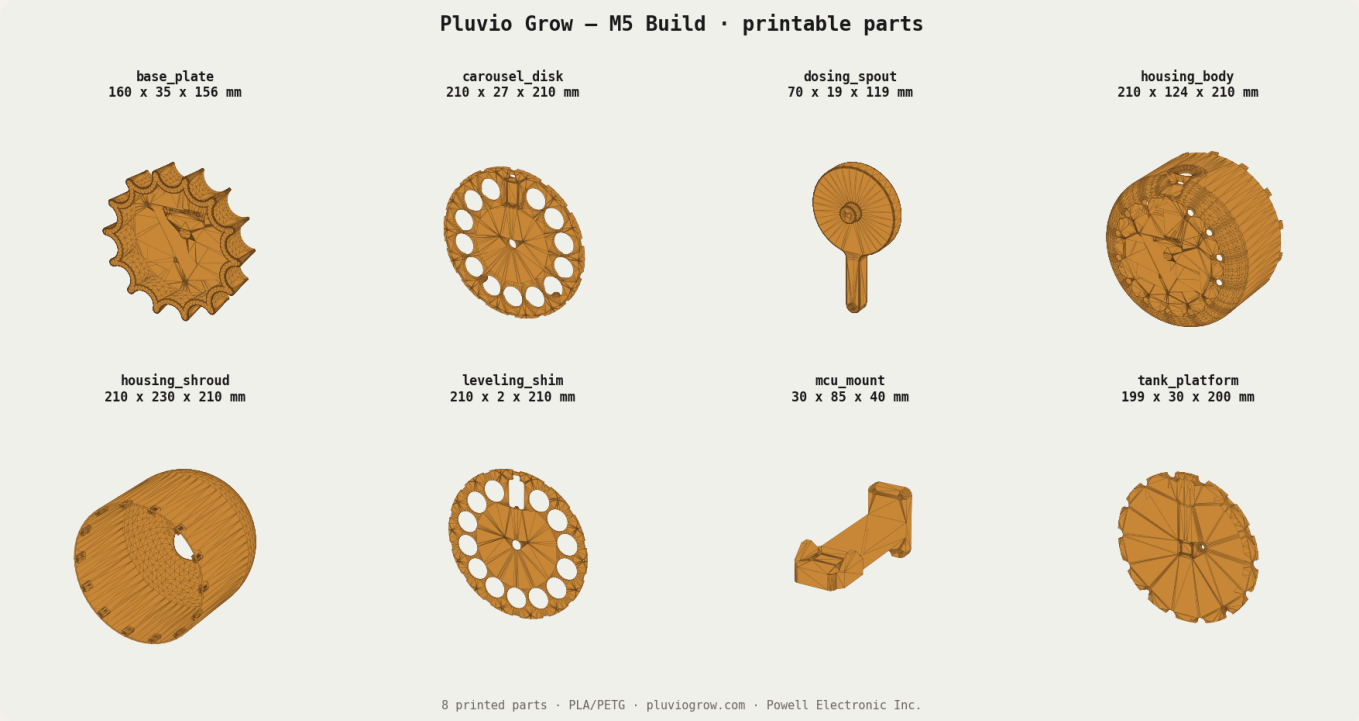
> you'll have spares.

## Mechanical & plumbing

| Part | Source | Notes |
|---|---|---|
| Solenoid valve, 5 V | AliExpress | the dosing valve |
| Solenoid driver (RC-PWM → DC) | AliExpress | lets the Atomic Motion servo channel switch the valve |
| **60 mL Luer-lock syringe** ×13 | generic (AliExpress / Amazon) | per-plant carousel vessels |
| **Luer-lock → 1/4″ tube adapter** ×13 | generic | connects the tube to each syringe |
| Silicone tubing, 1/8″ ID × 1/4″ OD, soft 50A | McMaster 51135K16 | one run per plant; ≈6 ft each, ~78 ft total for 13 — measure your layout and buy in bulk |
| 1-gallon utility jug | Uline S-10746B | reservoir |
| Top-hat rubber grommets (6 mm hole / 4 mm ID) | Amazon (uxcell 46-pc) | valve / tube seal |
| M3 × 8 mm thread-forming screws | McMaster 99461A941 | plastic assembly |
| M3 × 20 mm thread-forming screws | McMaster 99461A947 | plastic assembly |

## 3D-printed parts

Eight printed parts, included in `hardware/stl/`, from the confirmed M5 assembly (`hardware/step/pluvio_m5_assembly.STEP`). Print in PLA/PETG, 0.2 mm layers, 3–4 walls, 20–30% infill.




### Tools

Soldering iron, wire crimpers (for the Harwin sockets), 3D printer, small screwdriver, hobby knife, a cheap digital scale (to sanity-check the load cell).

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## Step 1 — Print the parts

Print all eight parts from `hardware/stl/`: `carousel_disk` (13 metering positions), `housing_body`, `housing_shroud`, `base_plate`, `tank_platform`, `dosing_spout`, `mcu_mount`, and `leveling_shim`. PLA or PETG, 0.2 mm layers, 3–4 walls, 20–30% infill. Most parts need no supports; orient the shroud and housing open-side-down. See `hardware/stl/README.md` for the full size table.

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## Step 2 — Wire the electronics

The **AtomS3R** sits on the **Atomic Motion Base**, which drives both the motor and the valve, so there's no separate motor driver board.

- **Encoder → Port B** and **Mini Scales → Port C** — these are **Grove** connectors: just plug the cable in, no soldering.
- **Stepper → Atomic Motion M1/M2** — convert to bipolar first (note below), then coil A → M1, coil B → M2 (crimp the Harwin sockets into the 2-pos housings).
- **Solenoid valve → driver module → "Servo 1"** — the firmware opens/closes the valve by writing servo angles to the driver.
- **Power** via the 5 V adapter / USB-C.

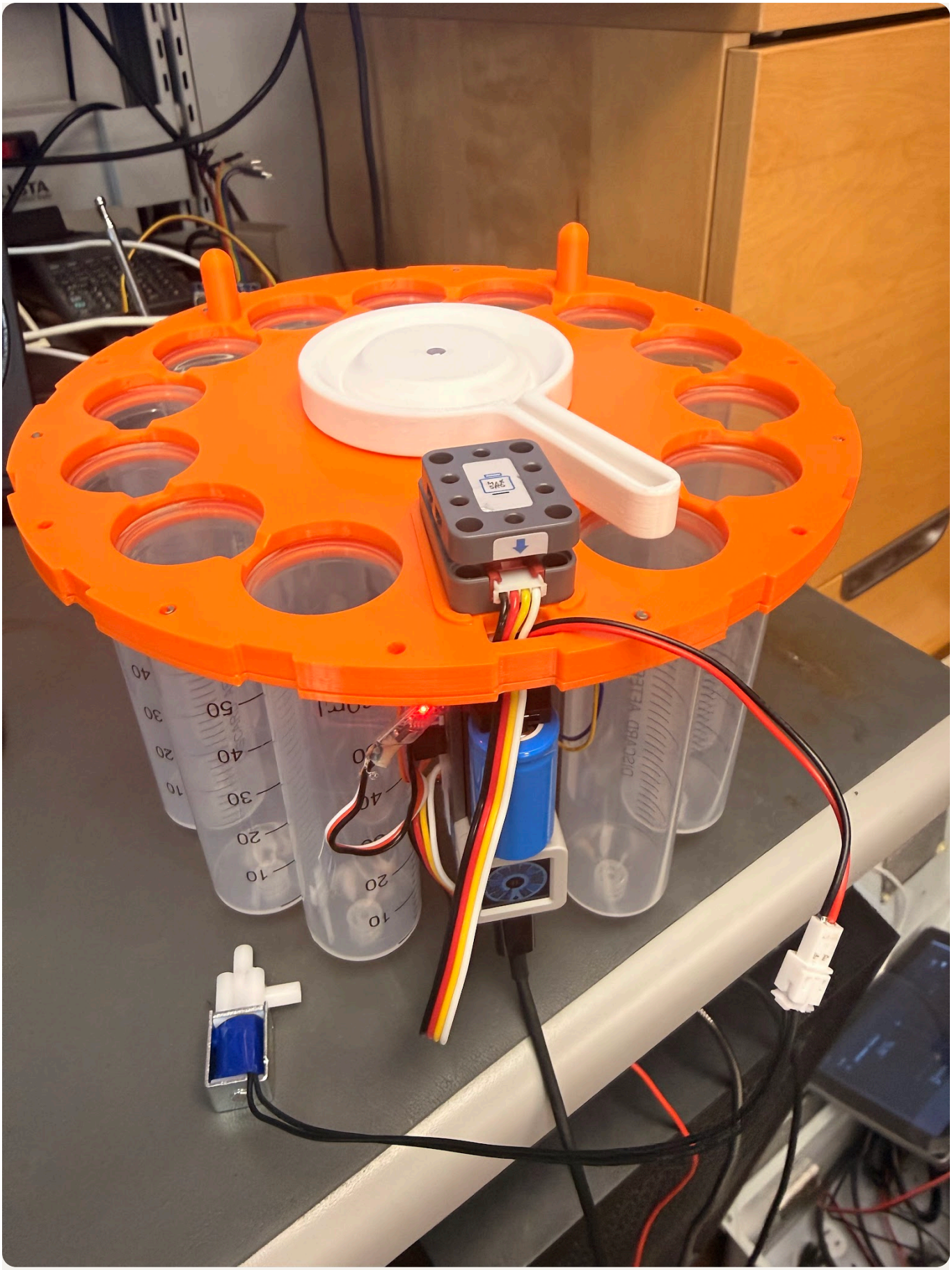
> **Bipolar mod (important):** the Olimex SM-5VDC ships as a stock **5-wire unipolar** 28BYJ-48. To drive it from the M5 H-bridge, convert it to a **4-wire bipolar** motor: open the motor and disconnect the shared common wire from the two internal coil pairs, so each coil is independent. Then coil A → M1, coil B → M2.

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## Step 3 — Assemble the mechanism

1. **Mount the MCU mount and the scale** (`mcu_mount` + Mini Scales) to the chassis with **two long (M3 × 20 mm) screws**.
2. **Seat the 13 syringes**, then drop the **shim** (`leveling_shim`) over them to hold them in place, and drive **every second short (M3 × 8 mm) screw** into the smaller-diameter holes.
3. **Mount the stepper motor** to the chassis with **two short screws**.
4. **Install the motor magnet into the dosing spout** (`dosing_spout`) and press the spout onto the **motor shaft** — this diametric magnet is what the AS5600 reads for arm position.
5. **Solder a cable header to the AS5600 encoder** and **glue it to the underside of the tank platform** (`tank_platform`). The encoder runs at **5 V**, with **DIR tied to GND**.
6. Once everything is **programmed and working**, **trim the 1-gallon jug** for a clearance fit inside the `housing_shroud`, and screw it down from the **bottom of the chassis with the longer screws**.

Seal the valve/tube passes with the top-hat grommets.



## Step 4 — Plumb it

Run one tube from the carousel spout to each plant; feed the pump/valve from the 1-gallon jug. Keep the reservoir at or below the unit so a closed valve can't siphon.

## Step 5 — Flash the firmware

Open-source, offline, no account. Full steps in `firmware/README.md`: install the ESP32 core + four libraries in the Arduino IDE, set the board, upload. On first boot it opens a `Pluvio-Setup-…` Wi-Fi hotspot — join it, pick your network, and manage the unit from its own web page.

## Step 6 — Calibrate & first water

Run the carousel calibration (it drives into both endstops to learn zero), set each plant's dose and interval, fill the jug, and let it run.

## Credits & license

Source-available from **Pluvio Grow**, a product of **Powell Electronic Inc.** Hardware & docs under **CC BY-NC-SA 4.0**; patent pending (US 64/071,683) on the weighed-dosing mechanism — personal/non-commercial builds welcome. Build one, show us, tag us. 🌱